Leo Marvin

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Profile

Mechanical engineering student at Arts et Métiers (ENSAM), France's leading engineering school for manufacturing and mechanics. My training combines workshop experience with technical project leadership on an AI-guided UAV developed for Dassault's UAV Challenge. Fluent in both English and French, with working knowledge of German.

Projects & Leadership Experience

Partnership with Pilion to design a new product September 2026 – Today

Pilion is a start-up incubated at Arts et Métiers working in construction. I'm heading the team of students recruited as part of one of our projects where we are developing BerSack.

- Responsible for developing BERSAC, a new product based on Pilion's needs
- Design and create a working prototype
- Building of a CAD model of the new prototype
- Conducting a design study aimed at sizing a functional prototype

Head of Sponsoring at Regatt'Art May 2026 – Today

The project aims to fully develop a boat to take part in the Monaco Energy Boat Challenge, a competition promoted by the Prince of Monaco: regattarts.fr

- Responsible for signing partnerships with companies
- Responsible for finding funds for the project (around 25 k€)
- In charge of handling communications with partners

Developing a model of an autonomous car May 2026

Project at Arts et Métiers for extra credit during the study of sensors

- Enabled study of signal processing with simultaneous processing of two video feeds
- Developed an experimental system allowing to study the impact of different parameters within the Python program

Head of the Dassault UAV Project Team representing Arts et Métiers Sept 2025 – May 2026

Although the team did not qualify after the second round, this was a very enriching project that produced a working drone

- Head of the Gadz'Air Team (12 members)
- Coordinated the 4 different sub-groups each tasked with a specific aspect of the drone (motor, fuselage, battery, information processing)
- Developed and assembled the Python code for the artificial intelligence video processing unit
  - Designing and writing 1st code which conducts **analysis of video feed to recognise objects** ;
  - Designing and writing 2st code which prevents **detects obstables and prevents crashes** (completes LiDAR)
  - **Built simulation** on Unreal Engine to test the Python codes
- Produced a detailed report for Dassault to prepare for the preselection (1st round of selections)
- Manufactured different prototypes for pieces of the drone to carry out tests
- Organised the transition to the second team

Research Project: The impact of the choice of materials on the production of eddy currents 2023 – 2025

- 2-year intensive research project related to eddy currents
- Developed an experimental system allowing to test and compare different disks, testing different types of steel with various geometries
- Produced a theoretical model and Python-based simulation
- Presented the project before a jury, received with honours (Awarded an A+)

Education

École Nationale Supérieure d'Arts et Métiers, Châlons-en-Champagne and Paris, France 2025 – 2028

One of France's leading Grande Ecole (Engineering School), 3 year program combined Bachelor and Master of Sciences

- Relevant courses:
  - Continuum Mechanics
  - Materials
  - Thermodynamics & Fluid Mechanics
  - Signal Processing
- Representing ENSAM at the Dassault UAV Challenge
- **GPA: 3.70**

Classe Préparatoire aux Grandes Écoles (CPGE PTSI & PT*), Lycée Raspail, Paris 2023 – 2025

Intensive, highly selective undergraduate program in mathematics, physics and engineering sciences, preparing for the national competitive entrance exams to France's top engineering schools (Grandes Écoles).

- Intensive courses in **mathematics**, **physics**, and **engineering sciences**
- Research Project: Development of a braking system relying on Eddy Currents
- Ranked in the top 10% (288 out of 3000)
- **GPA: 4.00**

Baccalauréat Général Option Internationale Britannique (Advanced Mathematics and Physics), Lycée Franco-Allemand, Buc, France 2020 – 2023

- **Graduated with honours** (within the top 20%)
- Half of classes taught in English
- Equivalent to a high school diploma with majors in **Mathematics** and **Physics** as well as minors **English Literature**, **English History** and **Geography**
- Part of the MUN (Model United Nations) debating club: participated in two MUN conferences including GrandMun with other high schools with bilingual sections

Professional Experience

Internship with Gris Découpage, Lesmenils, France June – August 2026

- Part of the logistics team (6 people)
- Reduced average delays in sorting by 30%

Seasonal worker with GAEC Chateau, Hermeray, France July – August 2025

- Part of a team of 3 people dealing with the maintenance of crops (3 hectares) as well as preparing for 2 markets each week

Charity Work

UNASS, Châlons-en-Champagne, France June 2026 – Today

- Part of the team setting up first aid centres for large events (up to 3000 people)
- Took part in 5 major events (more than 500 people)

Secretary of Gadz'Art Solidaires (Gasole), Châlons-en-Champagne, France June 2026 – Today

Charity organisation for Arts et Métiers, organises an annual music festival in Cluny (Le Grand Bastringue)

- Responsible for organising events (2 blood donations, partnership with the Tilleuls School in Fagnières to promote engineering and sciences in primary schools)
- Responsible for setting up a tutoring system with Compertrix school (2 sessions a week with 60 children)

Cordées de la Réussite September 2025 – July 2026

Working alongside ENSAM to promote engineering and science from primary to high school

- Responsible for organising workshops for 40 students

Charity work with Les Restos du Cœur (Food Bank), Rambouillet, France 2023 – Today

- Distribute food to 50 families weekly
- Took part in the campaign that produced a record local donation of 1.5 tons during the 39th National Food Campaign (national total: 9,000 tons)

Skills

- **Programming languages:** Proficient in Python (use of OpenCV and Chatbot development), LaTeX, HTML
- **Computer software / frameworks:** Microsoft Office Suite, SolidWorks, Fusion 360, Catia V5 and 3DExperience
- **Languages:** French (Native), English (Native – certified C2 by the French Ministry of Education – TOEFL :), German (Intermediate), Chinese (basics)
- **Hobbies:** Piano (5 years of classes and self-taught since), running (participated in 3 official 10K races and training for a half-marathon)